

SYLLABUS & WEIGHTAGE OF
QUESTIONS FOR COMPUTER BASED
TEST
ADVERTISEMENT No: 12/2026/CHQ/DR-CBT

MANAGER (ENGG.-ELECTRICAL)

Part-A

General Knowledge, General Intelligence,
General Aptitude, English etc.

Weightage 30%

Part-B

Questions on subjects relating to
Educational Qualifications

Weightage 70%

1. **Circuit Theory:** Circuit components; network graphs; KCL, KVL; circuit analysis methods; nodal analysis, mesh analysis; basic network theorems and applications; transient analysis; RL, RC and RLC circuits; sinusoidal steady state analysis; resonant circuits; coupled circuits; balanced 3-phase circuits; Two-port networks.
2. **Signals and systems:** Representation of continuous and discrete time signals; shifting and scaling operation; linear time invariant and causal systems; Fourier Laplace and Z transform.
3. **Instrumentation:** Insulation megger, earth megger, Kelvin's Double bridge, Quadrant electrometer, Rotating substandard, TOD meter.
4. **Electrical Machines:-**
 - a. **Transformers:** Constructional details – Principle of operation – vector diagrams on no load and load-regulation and efficiency – equivalent circuits and tests for the determination of parameters of equivalent circuits – types of three phase transformers and their applications – Scott connection of transformers.
 - b. **3-Phase Induction Motors:** Principle of operation – Cage and Slip ring motors – torque slip characteristics – methods of speed control.
 - c. **3-Phase Alternators:** Principle of operation and constructional details – types of Alternators – synchronous impedance – voltage regulation – short circuit ratio and its importance – phasor diagrams of round rotor and salient pole machines – synchronization – behaviour of an alternator connected to infinite bus – effect of varying excitation current and mechanical torque – power angle curves – control of active and reactive powers.
 - d. **3-Phase Synchronous Motors:** Principle of operation – torque developed and methods of starting – V and Inverted V curves – effects of variations of excitation – synchronous condensers.
5. **Single phase Induction Motors** – Types of single phase motors – Types of Single phase induction motors – characteristics and methods of starting – shaded pole induction motor.
6. **Transmission & Distribution:** Line constants – Inductance and Capacitance calculations –

Representation of overhead Lines – Short, Medium and Long lines – ABCD constants – Mechanical Design – Seg. Tension Calculations, Tuned Power Lines.

- a. Over Head Line Insulators:** Types of Insulators – Potential distributions over a string of suspensions insulators – string efficiency – Methods of improving string efficiency.
- b. Cable:** Insulation of cables – Grading of cables – Capacitance Measurement in cables – Testing of Cables – Power frequency withstand tests.
- c. Fault Calculations:** Balanced Fault calculations on systems – Symmetrical components – Types of faults – Analysis of unbalanced faults.
- d. Protection:** Characteristic of Relays – Over current, directional and distance protection of lines.
Protection of Alternators against stator faults, rotor faults, loss of excitation, unbalanced loading, overloading, failure of prime-mover. Over speeding and over voltage. Protection of transformers against winding faults, overloads and external short circuits.
- e. Circuit Breakers:** Air- blast, oil, minimum oil, vacuum – Sulphur hexafluoride and d.c. circuit breakers – Relative merits and demerits.

7. Power System Protection: Principles of over current, differential and distance protection, Concept of solid-state relays. Circuit breakers. Computer aided protection: Introduction; line bus, generator, transformer protection; numeric relays and application of DSP to protection.

8. Microprocessors and Microcomputers: PC organization; CPU, instruction set, register set, timing diagram, programming, interrupts, memory interfacing, I/O interfacing, programmable peripheral devices.

9. Analog and Digital Electronics: Characteristics of diodes. BJT, MOSFET, amplifiers – biasing, equivalent circuit and frequency response; oscillators and feedback amplifiers; operational amplifiers – characteristics and applications; simple active filters: VCOs and timers; combinational and sequential logic circuits; multiplexer; Schmitt trigger; multivibrators; sample and hold circuits; A/D and D/A converters; 8-bit microprocessor basics, architecture, programming and interfacing.

10. Power Electronics and Drives: Semiconductor power diodes, transistors thyristors, triacs, GTOs, MOSFETs and IGBTs – static characteristics and principles of operation; triggering circuits; phase control rectifiers; bridge converters – fully controlled and half controlled; principles of choppers and inverters; basic concepts of adjustable speed dc and ac drives.

11. Fiber Optic Systems Multiplexing: Time division multiplexing, frequency division multiplexing, optical properties of materials, refractive index absorption and emission of light, optical fibers lasers and optoelectronics materials, fiber optic lines.

12. Digital Communication: Pulse code modulation (PCM), differential pulse code modulation (DPCM), delta modulation (DM), Digital modulation and demodulation schemes: amplitude, phase and frequency keying schemes (ASK, PSK, FSK). Error control coding: error detection and correction, linear block codes, convolution codes. Information measure and source coding. Data networks, 7-layer architecture

13. Fundamentals of HVAC System and Various types of Air-conditioning equipment-

- Heating Plant- (Boilers/ Hot Water Generators)
- Central Air-conditioning – (Chillers-Centrifugal/Screw)
- VRV System, Precision Air conditioning
- Unitary A/c- window/split/cassette/Tower
- AHU, Cooling tower.

14. PUMPS

- a. PUMPS:** Water lifting devices, classification of pumps, centrifugal pump and its characteristics, specific speed, NPSH and cavitation, selection of pumps.
- b. HYDRAULICS AND FLUID MECHANICS:** Definition and properties of fluids. Units of measurement, kinematics of fluid flow, Bernoulli's equation and Euler's equation of motion. Dimensional analysis and similitude. Laminar and turbulent flow. General equation for head loss in pipes; open channel flow, hydraulic jump; Measurement of discharge in pipes and open channels.

15. Renewable energy sources i/c Solar power plant & its statutory guidelines.

ADDITIONAL TOPICS

1. Contract Management:

- Planning i/c estimation, tendering, execution, commissioning & operationalization.
- Project Planning, Scheduling & monitoring: Work breakdown structure, milestones, bar charts and critical path method basics.
- Execution of Electrical Works: Installation of MEP equipment, Site management, manpower planning, quality control, inspection procedures.
- Maintenance & Inventory management: Breakdown & Preventive Maintenance of MEP equipment i.e schedule, standard practice & maintaining records & electrical store.
- Understanding about Electrical safety codes i.e NBC, IS / IEC Standards, IE rules, Fire & safety norms, Environment norms and best practices etc

2. Sub-station

(i) HT Panels

- HT overhead Lines
- HT cables

(ii) Transformers

- LT panels, LT Cables
- Bus Duct, Capacitor panels
- Cable Trench, Cable Tray

3. DG Sets/UPS

- AMF Panel
- Synchronization
- SCADA & Power supply Management System

4. Lifts / Escalators

5. BMS/IBMS/EMS

- Building Management System
- Energy Management System

6. Energy Conservation Measurers

7. Water Supply

- Pumps Sets
- STP (Sewage Treatment Plant)
- WTP (Water Treatment Plant), RO Plant

8. Fire Alarm & Fire Fighting System

- Fire Alarm & Detection
- Wet Riser
- Down Comer
- Sprinkler System
- Fire Suppression System

9. Internal & External Electrification

- Wiring
- Conduiting
- Fitting/Fixtures
- Fans
- Pole (Street Light)
- High Mast Tower

10. Earthing & Lighting Protection System

11. CCTV / PA System
